

How to Convert an Angular App into a PWA

A progressive Web application (PWA) is a kind of application software that is delivered through the Web. In this short tutorial, let's go about converting an Angular application into a PWA.



progressive Web app (PWA) is a way to use modern browser features to create an app-like user experience for websites.

A PWA contains all the major features that native mobile applications do.

PWAs need to be:

Responsive: They should look fine on all devices with different resolutions.

Progressive: They should support all the latest browser features as well as old browsers.

Usable online: They should work without network connectivity.

Discoverable: They should get 'registered' so that they can be indexed by the browser.

Installable: One must be able to 'install' the Web app on a device, giving users an experience that is similar to using a native app.

The main goal of a PWA is to create a Web app with native app-like features such as a great UI, offline support, etc.

If you are reading this, you probably are a Web developer, in which case, I

will assume that you also have Google Chrome installed.

Auditing your application using Lighthouse

While you are on this page in Chrome, open Developer Tools (F12 on the keyboard). Flip over to the 'Audits' tab. Depending on the version of Chrome, you'll see a few different features to run audits against. Choose 'Desktop' for the device, and select only 'Progressive Web App' from the 'Audits' list.

Click on 'Run audits' and wait a few seconds to see how well *Alligator.io* does as a PWA! You can run the Lighthouse audit tool on any page to see how well it implements the aspects of a PWA.

Make your Angular application a PWA

Let's go ahead and create a sample Angular application. You can see how easy it is to convert it to a PWA. From a terminal, with the Angular CLI installed globally, run the following command:

```
$ ng new pwa-example
```

This just sets up a default, out-of-the-box, 'Hello World' Angular application. Alternatively, generate a new project adding any properties that you like, or run the next command in an existing Angular project.

Change directories to your new project folder, and run the following command:

```
$ ng add @angular/pwa && ng build --prod
```

...and just like that, your Angular application is production-ready.

Let's look at some important changes that occurred:

- *angular.json* was modified to include the *serviceWorker: true* property to the build configuration, as well as specifying that *src/manifest.webmanifest* be included as a build asset.
- *manifest.webmanifest* is the Web